Practice workbook · Showing a cubic has only one real root

Proving that a cubic equation has exactly one real root by showing its quadratic factor has a negative discriminant.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-IDN-12

1. Question 4 · part b · 3 marks

2022 JUN · Paper 02 (regional R)

Worked-example retry · Attempt part b only.

4 $f(x) = x^3 + px^2 + qx + 7$ where p and q are integers.

$(x + 1)$ is a factor of $f(x)$
The remainder when $f(x)$ is divided by $(x + 2)$ is -5

(a) Find the value of p and the value of q

(5)

(b) Hence, show that $f(x) = 0$ has only one real root.

(3)

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P 7 1 6 4 2 A 0 1 0 3 6

1. Question 4 · part b · 3 marks

2022 JUN · Paper 02 (regional R)

Worked-example retry · Attempt part b only.

Question 4 continued

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(Total for Question 4 is 8 marks)

1. Question 4 · part b · 3 marks · Working and answer

Working and answer

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2. Question 5 · part b · 4 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

5 $f(x) = x^3 + x^2 + x + c$ where c is a constant

The remainder when $f(x)$ is divided by $(x-2)$ is 4 times the remainder when $f(x)$ is divided by $(x+1)$

(a) Show that $c = 6$

(4)

Given that $(x+2)$ is a factor of $f(x)$

(b) show that the equation $f(x) = 0$ has only one real root.

(4)

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P 7 6 5 0 9 A 0 1 2 3 2

2. Question 5 · part b · 4 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

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Question 5 continued

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(Total for Question 5 is 8 marks)

2. Question 5 · part b · 4 marks · Working and answer

Working and answer

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3. Question 1 · part b · 4 marks

2024 JUN · Paper 02

Further practice · Attempt part b only.

1 $f(x) = 6x^3 - 13x^2 + ax - 10$ where a is a constant

Given that $(3x - 2)$ is a factor of $f(x)$

(a) show that $a = 21$

(2)

(b) Hence show algebraically that the curve $y = f(x)$ has only one intersection with the x-axis.

(4)

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(Total for Question 1 is 6 marks)

3. Question 1 · part b · 4 marks · Working and answer

Working and answer

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4. Question 10 · part d · 4 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part d only.

10 (a) Use the factor theorem to show that $(4x - 3)$ is a factor of

$$16x^3 + 11x - 15$$

(2)

(b) Using formulae given on page 2, show that

(i) $\sin 2\theta = 2\sin\theta\cos\theta$

(ii) $\cos 2\theta = 2\cos^2\theta - 1$

(5)

(c) Show that the equation

$$27\cos\theta\cos 2\theta + 19\sin\theta\sin 2\theta - 15 = 0$$

becomes the equation

$$16x^3 + 11x - 15 = 0$$

by using the substitution $x = \cos\theta$

(4)

(d) Hence show that any solution of the equation

$$27\cos\theta\cos 2\theta + 19\sin\theta\sin 2\theta - 15 = 0$$

is given by $\cos\theta = \frac{3}{4}$

(4)

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4. Question 10 · part d · 4 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part d only.

Question 10 continued

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4. Question 10 · part d · 4 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part d only.

Question 10 continued

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4. Question 10 · part d · 4 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part d only.

Question 10 continued

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(Total for Question 10 is 15 marks)

4. Question 10 · part d · 4 marks · Working and answer

Working and answer

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5. Question 1 · part b · 5 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

1 $f(x) = (x - 3)[x^2 + (p - 2)x + q]$

Given that $f(0) = -12$

(a) find the value of q.

(2)

(b) Find the range of values of p for which the cubic equation $f(x) = 0$ has only one real root.

(5)

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(Total for Question 1 is 7 marks)

5. Question 1 · part b · 5 marks · Working and answer

Working and answer

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